

## EEE 312 Semester Test, 2023/2024 Session

- 1) State the reason why BPSK modulation is sometimes referred to as PRK and bi-phase modulation
- 2) Show that the channel capacity of an ideal AWGN channel with infinite bandwidth is given by  $C_{\infty} = 1.44 \frac{S}{N_0}$
- 3) How can 11100010 code word be represented using the following line coding methods?
  - (i) Unipolar RZ
  - (ii) Unipolar NRZ-L
  - (iii) Bipolar RZ

$S$  = sig power.

$N_0$  = Power spectr densit  
white noise



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SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY  
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING  
2<sup>ND</sup> SEMESTER B.ENG. EXAMINATION 2023/2024 SESSION  
EEE 312 DIGITAL COMMUNICATION PRINCIPLES

INSTRUCTIONS: Answer any four (4) questions. Carefully read all questions, provide diagrams where needed. Course Unit: 2 units. Time Allowed: 2 hrs.

**Question 1**

- (i) Explain digital communication system with a clear diagram (8 marks)
- (ii) Use waveform to show how digital information can be conveyed using the three major band pass digital modulation. **Note:** Use binary bit of your choice. (3 marks)
- (iii) Briefly discuss the significance of SNR in communication system (2 marks)
- (iv) A discrete source emits one of five symbols once every milliseconds with probabilities 0.500, 0.250, 0.125, 0.0625, and 0.5 respectively. Determine the source entropy. (2 marks)

**Question 2**

- (i) State the reason why BPSK modulation is sometimes referred to as PRK and bi-phase modulation (2 marks)
- (ii) Show that the channel capacity of an ideal AWGN channel with infinite bandwidth is given by

$$C_{\infty} = 1.44 \frac{S}{\eta}$$

where  $S$  is the average signal power and  $\eta$  is the PSD of white Gaussian noise. (6 marks)

- (iii) The bandwidth of an input signal to a PCM is restricted to 4 kHz. The input signal varies in amplitude from -3.8 V to 3.8 V and has the average power of 30 mW. The required signal to noise ratio is given as 20 dB. The PCM modulator produces binary output. Assume uniform quantization, estimate

- (a) The number of bit required per sample. (3 marks)
- (b) If the input of 30 of such PCM coders are time multiplexed, what would be the minimum required transmission bandwidth for the multiplexed signals. (4 marks)

**Question 3**

- (i) State sampling theorem. (2 marks)
- (ii) From the first principle, prove that the normalized signal to quantization noise ratio in dB is expressed as

$$\left( \frac{S}{N} \right) \leq (4.8 + 6v) \quad (6 \text{ marks})$$

- (iii) An analog signal having 4 kHz bandwidth is sampled at 1.25 times the Nyquist rate and each sample is quantized into one of equally likely levels. Assume that successive samples are statistically independent.

- (a) What is the information rate of the source? (3 marks)



$$C = B \log_2(1 + S/N)$$

$$10 \text{ kHz} = 8 \text{ [?]}$$

- (b) Can the output of the source be transmitted without error on an AWGN channel with bandwidth of 10 KHz at an S/N ratio of 20 dB? (4 marks)

#### Question 4

- (i) Using the constellation diagram in Fig. Q4(i), answer the following questions:

(a) What type of modulation does this represent; (1 mark)

(b) How many symbols are represented; and (1 mark)

(c) Determine the number of bits per symbols used? (1 mark)

- (ii) Sketch the constellation diagram of the following digital modulation scheme;

(a) OOK (1 mark) *unrated*

(b) 8-PSK (1 mark)

(c) 16-QAM (1 mark)

(d) OQPSK. (1 mark) *unrated*

- (iii) Binary data is transmitted over a microwave link at a rate of  $10^6$  bits/seconds and power spectral density of noise at the receiver input is  $10^{-10}$  W/Hz. Find the average carrier power required to maintain an average probability of error  $P_e \leq 10^{-4}$  for coherent BFSK system. (8 marks)

$$C = B \log_2(1 + S/N)$$

$$S/N_{dB} = 4.8 + 6V \rightarrow V = 3$$

$$S/N = 3 \times 2^{2V} = \boxed{192} = C$$

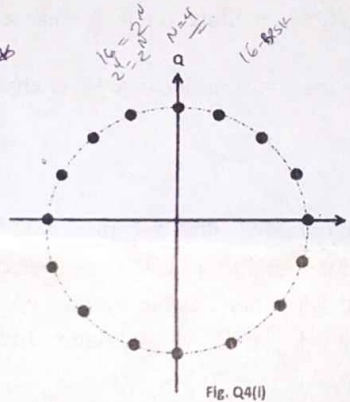


Fig. Q4(i)

$$2^4 = 2^2 = 16$$

$$8 = 2^3$$

#### Question 5

- (i) State 3 advantages of spread spectrum technique. (3 marks)

- (ii) With the aid of a well labeled diagram and associated expressions, discuss direct sequence spread spectrum technique (6 marks)

- (iii) How can 1110001011 code word be represented using the following line coding methods?

(a) Unipolar RZ (1 mark)

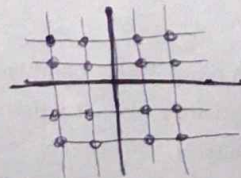
(b) Unipolar NRZ-S (1 mark)

(c) Unipolar NRZ-L (1 mark)

(d) Bipolar RZ (1 mark)

(e) Polar NRZ (1 mark)

(f) Split- $\phi$  Manchester coding (1 mark)



|      |    |
|------|----|
| 0000 | 0  |
| 0001 | 1  |
| 0010 | 2  |
| 0011 | 3  |
| 0100 | 4  |
| 0101 | 5  |
| 0110 | 6  |
| 0111 | 7  |
| 1000 | 8  |
| 1001 | 9  |
| 1010 | 10 |
| 1011 | 11 |
| 1100 | 12 |
| 1101 | 13 |
| 1110 | 14 |
| 1111 | 15 |

$$2/16$$

$$\frac{2}{2} \frac{8}{2} \frac{0}{2} \frac{0}{2} \frac{0}{2} \frac{1}{2} \frac{0}{2} \frac{1}{2}$$



Panel with  
marks)

Complementary Error Function Table

| x    | erfc(x)  | x    | erfc(x)  | x    | erfc(x)  | x    | erfc(x)  | x    | erfc(x)  | x    | erfc(x)  | x    | erfc(x)    |
|------|----------|------|----------|------|----------|------|----------|------|----------|------|----------|------|------------|
| 0    | 1.000000 | 0.5  | 0.479500 | 1    | 0.157299 | 1.5  | 0.033895 | 2    | 0.004678 | 2.5  | 0.000407 | 3    | 0.00002209 |
| 0.01 | 0.988717 | 0.51 | 0.470756 | 1.01 | 0.153190 | 1.51 | 0.032723 | 2.01 | 0.004475 | 2.51 | 0.000386 | 3.01 | 0.00002074 |
| 0.02 | 0.977435 | 0.52 | 0.462101 | 1.02 | 0.149162 | 1.52 | 0.031587 | 2.02 | 0.004281 | 2.52 | 0.000365 | 3.02 | 0.00001947 |
| 0.03 | 0.966159 | 0.53 | 0.453536 | 1.03 | 0.145216 | 1.53 | 0.030484 | 2.03 | 0.004094 | 2.53 | 0.000346 | 3.03 | 0.00001827 |
| 0.04 | 0.954889 | 0.54 | 0.445061 | 1.04 | 0.141350 | 1.54 | 0.029414 | 2.04 | 0.003914 | 2.54 | 0.000328 | 3.04 | 0.00001714 |
| 0.05 | 0.943628 | 0.55 | 0.436677 | 1.05 | 0.137564 | 1.55 | 0.028377 | 2.05 | 0.003742 | 2.55 | 0.000311 | 3.05 | 0.00001608 |
| 0.06 | 0.932378 | 0.56 | 0.428384 | 1.06 | 0.133856 | 1.56 | 0.027372 | 2.06 | 0.003577 | 2.56 | 0.000294 | 3.06 | 0.00001508 |
| 0.07 | 0.921142 | 0.57 | 0.420184 | 1.07 | 0.130227 | 1.57 | 0.026397 | 2.07 | 0.003418 | 2.57 | 0.000278 | 3.07 | 0.00001414 |
| 0.08 | 0.909922 | 0.58 | 0.412077 | 1.08 | 0.126674 | 1.58 | 0.025453 | 2.08 | 0.003266 | 2.58 | 0.000264 | 3.08 | 0.00001326 |
| 0.09 | 0.898719 | 0.59 | 0.404064 | 1.09 | 0.123197 | 1.59 | 0.024538 | 2.09 | 0.003120 | 2.59 | 0.000249 | 3.09 | 0.00001243 |
| 0.1  | 0.887537 | 0.6  | 0.396144 | 1.1  | 0.119795 | 1.6  | 0.023652 | 2.1  | 0.002979 | 2.6  | 0.000236 | 3.1  | 0.00001165 |
| 0.11 | 0.876377 | 0.61 | 0.388319 | 1.11 | 0.116467 | 1.61 | 0.022793 | 2.11 | 0.002845 | 2.61 | 0.000223 | 3.11 | 0.00001092 |
| 0.12 | 0.865242 | 0.62 | 0.380589 | 1.12 | 0.113212 | 1.62 | 0.021962 | 2.12 | 0.002716 | 2.62 | 0.000211 | 3.12 | 0.00001023 |
| 0.13 | 0.854133 | 0.63 | 0.372954 | 1.13 | 0.110029 | 1.63 | 0.021157 | 2.13 | 0.002593 | 2.63 | 0.000200 | 3.13 | 0.00000958 |
| 0.14 | 0.843053 | 0.64 | 0.365414 | 1.14 | 0.106918 | 1.64 | 0.020378 | 2.14 | 0.002475 | 2.64 | 0.000189 | 3.14 | 0.00000897 |
| 0.15 | 0.832004 | 0.65 | 0.357971 | 1.15 | 0.103876 | 1.65 | 0.019624 | 2.15 | 0.002361 | 2.65 | 0.000178 | 3.15 | 0.00000840 |
| 0.16 | 0.820988 | 0.66 | 0.350623 | 1.16 | 0.100904 | 1.66 | 0.018895 | 2.16 | 0.002253 | 2.66 | 0.000169 | 3.16 | 0.00000786 |
| 0.17 | 0.810008 | 0.67 | 0.343372 | 1.17 | 0.098000 | 1.67 | 0.018190 | 2.17 | 0.002149 | 2.67 | 0.000159 | 3.17 | 0.00000736 |
| 0.18 | 0.799064 | 0.68 | 0.336218 | 1.18 | 0.095163 | 1.68 | 0.017507 | 2.18 | 0.002049 | 2.68 | 0.000151 | 3.18 | 0.00000689 |
| 0.19 | 0.788160 | 0.69 | 0.329160 | 1.19 | 0.092392 | 1.69 | 0.016847 | 2.19 | 0.001954 | 2.69 | 0.000142 | 3.19 | 0.00000644 |
| 0.2  | 0.777297 | 0.7  | 0.322199 | 1.2  | 0.089686 | 1.7  | 0.016210 | 2.2  | 0.001863 | 2.7  | 0.000134 | 3.2  | 0.00000603 |
| 0.21 | 0.766478 | 0.71 | 0.315335 | 1.21 | 0.087045 | 1.71 | 0.015593 | 2.21 | 0.001776 | 2.71 | 0.000127 | 3.21 | 0.00000564 |
| 0.22 | 0.755704 | 0.72 | 0.308567 | 1.22 | 0.084466 | 1.72 | 0.014997 | 2.22 | 0.001692 | 2.72 | 0.000120 | 3.22 | 0.00000527 |
| 0.23 | 0.744977 | 0.73 | 0.301896 | 1.23 | 0.081950 | 1.73 | 0.014422 | 2.23 | 0.001612 | 2.73 | 0.000113 | 3.23 | 0.00000493 |
| 0.24 | 0.734300 | 0.74 | 0.295322 | 1.24 | 0.079495 | 1.74 | 0.013865 | 2.24 | 0.001536 | 2.74 | 0.000107 | 3.24 | 0.00000460 |
| 0.25 | 0.723674 | 0.75 | 0.288845 | 1.25 | 0.077100 | 1.75 | 0.013328 | 2.25 | 0.001463 | 2.75 | 0.000101 | 3.25 | 0.00000430 |
| 0.26 | 0.713100 | 0.76 | 0.282463 | 1.26 | 0.074764 | 1.76 | 0.012810 | 2.26 | 0.001393 | 2.76 | 0.000095 | 3.26 | 0.00000402 |
| 0.27 | 0.702582 | 0.77 | 0.276179 | 1.27 | 0.072486 | 1.77 | 0.012309 | 2.27 | 0.001326 | 2.77 | 0.000090 | 3.27 | 0.00000376 |
| 0.28 | 0.692120 | 0.78 | 0.269990 | 1.28 | 0.070266 | 1.78 | 0.011826 | 2.28 | 0.001262 | 2.78 | 0.000084 | 3.28 | 0.00000351 |
| 0.29 | 0.681717 | 0.79 | 0.263897 | 1.29 | 0.068101 | 1.79 | 0.011359 | 2.29 | 0.001201 | 2.79 | 0.000080 | 3.29 | 0.00000328 |
| 0.3  | 0.671373 | 0.8  | 0.257899 | 1.3  | 0.065992 | 1.8  | 0.010909 | 2.3  | 0.001143 | 2.8  | 0.000075 | 3.3  | 0.00000306 |
| 0.31 | 0.661092 | 0.81 | 0.251997 | 1.31 | 0.063937 | 1.81 | 0.010475 | 2.31 | 0.001088 | 2.81 | 0.000071 | 3.31 | 0.00000285 |
| 0.32 | 0.650874 | 0.82 | 0.246189 | 1.32 | 0.061935 | 1.82 | 0.010057 | 2.32 | 0.001034 | 2.82 | 0.000067 | 3.32 | 0.00000266 |
| 0.33 | 0.640721 | 0.83 | 0.240476 | 1.33 | 0.059985 | 1.83 | 0.009653 | 2.33 | 0.000984 | 2.83 | 0.000063 | 3.33 | 0.00000249 |
| 0.34 | 0.630635 | 0.84 | 0.234857 | 1.34 | 0.058086 | 1.84 | 0.009264 | 2.34 | 0.000935 | 2.84 | 0.000059 | 3.34 | 0.00000232 |
| 0.35 | 0.620618 | 0.85 | 0.229332 | 1.35 | 0.056238 | 1.85 | 0.008889 | 2.35 | 0.000889 | 2.85 | 0.000056 | 3.35 | 0.00000216 |
| 0.36 | 0.610670 | 0.86 | 0.223900 | 1.36 | 0.054439 | 1.86 | 0.008528 | 2.36 | 0.000845 | 2.86 | 0.000052 | 3.36 | 0.00000202 |
| 0.37 | 0.600794 | 0.87 | 0.218560 | 1.37 | 0.052688 | 1.87 | 0.008179 | 2.37 | 0.000803 | 2.87 | 0.000049 | 3.37 | 0.00000188 |
| 0.38 | 0.590991 | 0.88 | 0.213313 | 1.38 | 0.050984 | 1.88 | 0.007844 | 2.38 | 0.000763 | 2.88 | 0.000046 | 3.38 | 0.00000175 |
| 0.39 | 0.581261 | 0.89 | 0.208157 | 1.39 | 0.049327 | 1.89 | 0.007521 | 2.39 | 0.000725 | 2.89 | 0.000044 | 3.39 | 0.00000163 |
| 0.4  | 0.571608 | 0.9  | 0.203092 | 1.4  | 0.047715 | 1.9  | 0.007210 | 2.4  | 0.000689 | 2.9  | 0.000041 | 3.4  | 0.00000152 |
| 0.41 | 0.562031 | 0.91 | 0.198117 | 1.41 | 0.046148 | 1.91 | 0.006910 | 2.41 | 0.000654 | 2.91 | 0.000039 | 3.41 | 0.00000142 |
| 0.42 | 0.552532 | 0.92 | 0.193232 | 1.42 | 0.044624 | 1.92 | 0.006622 | 2.42 | 0.000621 | 2.92 | 0.000036 | 3.42 | 0.00000132 |
| 0.43 | 0.543113 | 0.93 | 0.188437 | 1.43 | 0.043143 | 1.93 | 0.006344 | 2.43 | 0.000589 | 2.93 | 0.000034 | 3.43 | 0.00000123 |
| 0.44 | 0.533775 | 0.94 | 0.183729 | 1.44 | 0.041703 | 1.94 | 0.006077 | 2.44 | 0.000559 | 2.94 | 0.000032 | 3.44 | 0.00000115 |
| 0.45 | 0.524518 | 0.95 | 0.179109 | 1.45 | 0.040305 | 1.95 | 0.005821 | 2.45 | 0.000531 | 2.95 | 0.000030 | 3.45 | 0.00000107 |
| 0.46 | 0.515345 | 0.96 | 0.174576 | 1.46 | 0.038946 | 1.96 | 0.005574 | 2.46 | 0.000503 | 2.96 | 0.000028 | 3.46 | 0.00000099 |
| 0.47 | 0.506255 | 0.97 | 0.170130 | 1.47 | 0.037627 | 1.97 | 0.005336 | 2.47 | 0.000477 | 2.97 | 0.000027 | 3.47 | 0.00000092 |
| 0.48 | 0.497250 | 0.98 | 0.165769 | 1.48 | 0.036346 | 1.98 | 0.005108 | 2.48 | 0.000453 | 2.98 | 0.000025 | 3.48 | 0.00000086 |
| 0.49 | 0.488332 | 0.99 | 0.161492 | 1.49 | 0.035102 | 1.99 | 0.004889 | 2.49 | 0.000429 | 2.99 | 0.000024 | 3.49 | 0.00000080 |

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